応用数学　3章　第9回　Basic 解答

【Basic】

問題 142

次の関数のフーリエ級数を求めよ．(1) $f(x) = \begin{cases} 2 & (-1 \leq x < 0) \\ 1 & (0 \leq x < 1) \end{cases}$，$f(x+2) = f(x)$ (2) $g(x) = \begin{cases} 3 & (-3 \leq x < 0) \\ x & (0 \leq x < 3) \end{cases}$，$g(x+6) = g(x)$ [教 p.82(問 3)]

解答:

(1) $a_0 = \dfrac{1}{2}\displaystyle\int_{-1}^{0} 2\,dx + \dfrac{1}{2}\int_{0}^{1} 1\,dx = \dfrac{3}{2}$

$a_n = \displaystyle\int_{-1}^{0} 2\cos n\pi x\,dx + \int_{0}^{1} \cos n\pi x\,dx = 0$

$b_n = \displaystyle\int_{-1}^{0} 2\sin n\pi x\,dx + \int_{0}^{1} \sin n\pi x\,dx = \dfrac{2((-1)^n-1)}{n\pi} + \dfrac{1-(-1)^n}{n\pi}$

$= \dfrac{(-1)^n-1}{n\pi}$

$f(x) = \dfrac{3}{2} - \dfrac{2}{\pi}\displaystyle\sum_{k=0}^{\infty} \dfrac{\sin(2k+1)\pi x}{2k+1}$

(2) 周期 $2l = 6$，$l = 3$ として $a_0 = \dfrac{1}{6}\displaystyle\int_{-3}^{0} 3\,dx + \dfrac{1}{6}\int_{0}^{3} x\,dx = \dfrac{3}{2} + \dfrac{3}{4} = \dfrac{9}{4}$

$a_n = \dfrac{1}{3}\displaystyle\int_{-3}^{0} 3\cos\dfrac{n\pi x}{3}\,dx + \dfrac{1}{3}\int_{0}^{3} x\cos\dfrac{n\pi x}{3}\,dx = \dfrac{3((-1)^n-1)}{n^2\pi^2}$

$b_n = \dfrac{1}{3}\displaystyle\int_{-3}^{0} 3\sin\dfrac{n\pi x}{3}\,dx + \dfrac{1}{3}\int_{0}^{3} x\sin\dfrac{n\pi x}{3}\,dx = -\dfrac{3}{n\pi}$

$g(x) = \dfrac{9}{4} - \dfrac{6}{\pi^2}\displaystyle\sum_{k=0}^{\infty} \dfrac{1}{(2k+1)^2}\cos\dfrac{(2k+1)\pi x}{3} - \dfrac{3}{\pi}\sum_{n=1}^{\infty} \dfrac{1}{n}\sin\dfrac{n\pi x}{3}$